$\mathbb{R}^4 = \mathbb{R} \times \mathbb{R}^3$

<u>path integral approach</u> $\qquad \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - m^2 \phi^2$

$$S[\phi] = \int_{\mathbb{R}^4} d^4x \, \mathcal{L}(\phi, \partial\phi)$$

$\phi : \mathbb{R}^4 \to \mathbb{R}$

$$Z = \int D\phi \, e^{iS[\phi]}$$

translation invariant

<u>2 problems:</u> ① there is no Lebesgue measure on set of all fields

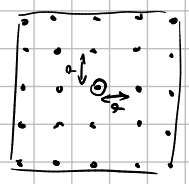
② exponential oscillates

---

to solve problem ① we can discretize spacetime

we lattice $\Lambda = \left\{ \sum_\mu a \lambda^\mu e_\mu , \quad \lambda^\mu = -N, -N+1, \dots, N-1, N \right\} \qquad a = \frac{L}{N}$

$= \{ x_i \}_{i \in I}$



$$\hookrightarrow Z = \prod_i \int_{\mathbb{R}} d\phi_i \, e^{iS[\phi]}$$

② "solve" it by doing a Wick rotation: $t \to it$

$$i \int_{\mathbb{R}^4} d^4x \, \frac{1}{2} \left((\partial_t \phi)^2 - (\nabla\phi)^2 \right)$$

$dt \to i\,dt$

$\frac{\partial}{\partial t} \to -i \frac{\partial}{\partial t}$

$$\to \quad i^2 \int_{\mathbb{R}^4} d^4x \, \frac{1}{2} \left(-(\partial_t \phi)^2 - (\nabla\phi)^2 \right)$$

$$= \frac{1}{2} \int_{\mathbb{R}^4} d^4x \left((\partial_t \phi)^2 + (\nabla\phi)^2 \right)$$

integration by parts

$$= -\frac{1}{2} \int_{\mathbb{R}^4} d^4x \left(\phi \partial_t^2 \phi + \phi \nabla^2 \phi \right)$$

$\nabla \phi \cdot \nabla \phi = \nabla \cdot (\phi \nabla \phi) - \phi \nabla^2 \phi$

$$e^{-\frac{1}{2}\int_{\mathbb{R}^4} d^4x \, (\phi L \phi)}$$ $L = \partial_t^2 + \nabla^2$ linear differential operator

→ probability measure

multi-dimensional Gaussian on $\mathbb{R}^n$

$$e^{-\frac{1}{2} x^t A x}$$ A matrix (symmetric + pos. definite)

<u>Gaussian distribution</u>: (1D) $p(x) = e^{-\frac{1}{2}x^2}$

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} dx \, e^{-\frac{1}{2}x^2} f(x)}{\int_{-\infty}^{\infty} dx \, e^{-\frac{1}{2}x^2}}$$

nD version $p(\vec{x}) = e^{-\frac{1}{2} x^t A x}$

$$\langle x^i \rangle = \frac{\int_{\mathbb{R}^n} d^n x \, e^{-\frac{1}{2}x^t A x} x^i}{\int_{\mathbb{R}^n} d^n x \, e^{-\frac{1}{2}x^t A x}}$$

$$\langle \phi(x) \rangle = \frac{\int D\phi \, e^{S[\phi]} \phi(x)}{\int D\phi \, e^{S[\phi]}} = 0$$ (1-point function)

$x \in \mathbb{R}^4$

quantization step

(interpretation of $\langle \phi(x^1) \phi(x^2) \phi(x^3) \dots \rangle$

$$\langle 0 | \hat{\phi}(t, \vec{x}) | 0 \rangle$$

<u>Theorem</u>: <u>Isserlis theorem</u>

For a gaussian distribution $p(x) = e^{-\frac{1}{2} x^t A x}$ $x \in \mathbb{R}^n$

$$\langle x^{i_1} x^{i_2} \dots x^{i_k} \rangle = \frac{\int_{\mathbb{R}^n} d^n x \, p(x) \, x^{i_1} \dots x^{i_k}}{\int_{\mathbb{R}^n} d^n x \, p(x)}$$

$$= \sum_{\sigma \in P_k^2} \prod_{\{r,s\}\in\sigma} \langle x^{i_r} x^{i_s} \rangle$$

P_k^2 set of all partitions into pairs of $\{1, \dots, k\}$

<u>Corollary:</u> $= 0$ if k is odd

<u>ex:</u> $k = 4$ $\{1, 2, 3, 4\}$

<u>partitions:</u> $(1,2)\ (3,4)$

$(1,3)\ (2,4)$

$(1,4)\ (2,3)$

$$\langle x^{i_1} \dots x^{i_4} \rangle = \langle x^{i_1} x^{i_2} \rangle \langle x^{i_3} x^{i_4} \rangle + \langle x^{i_1} x^{i_3} \rangle \langle x^{i_2} x^{i_4} \rangle + \langle x^{i_1} x^{i_4} \rangle \langle x^{i_2} x^{i_3} \rangle$$

<u>for path integral:</u>

$$\langle \phi(x^{i_1}) \dots \phi(x^{i_k}) \rangle = \sum_{\sigma \in P_k^2} \prod_{\{r,s\}\in\sigma} \langle \phi(x^{i_r}) \phi(x^{i_s}) \rangle$$

<u>figure out the 2-point function</u> $\underbrace{\langle 0 | \hat{\phi}(x)}\ \underbrace{\hat{\phi}(x') | 0 \rangle}$

<u>for a Gaussian</u>

$$\langle x^i x^j \rangle = \frac{\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2} x^t A x}\, x^i x^j}{\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2} x^t A x}} = (A^{-1})_{ij}$$

<u>for path integrals</u> $\langle \phi(x) \phi(x') \rangle := (L^{-1})(x, x')$

$$P(\phi) = e^{-\frac{1}{2} \int_{\mathbb{R}^4} d^4 x\, (\phi L \phi)}$$

$L^{-1}(x, x') = G(x, x')$ s.t $\underline{L G(x, x') = \delta(x - x')}$ Green's function

in our case $L = \partial_t^2 + \nabla^2$ $L^{-1} = \dfrac{1}{\partial_t^2 + \nabla^2}$

FT

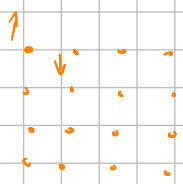
$(k_0^2 + \vec{k}^2)\, \hat{G}(k) = 1$

<u>Fourier transform:</u> $\dfrac{1}{k_0^2 + \vec{k}^2}$

in QM can do the same

$$Z[q] = \int Dq \, e^{\frac{i}{\hbar} S[q]}$$ "sum over all paths"

Statistical mechanics $$Z = \sum_i e^{-\beta E_i}$$



Connection (path integral $\leftrightarrow$ canonical quantization)

$$\boxed{\langle \phi(\underbrace{x_1}_{\in \mathbb{R}^4}) \phi(x_2) \cdots \phi(x_n) \rangle \equiv \langle 0 | T\big(\hat{\phi}(\overbrace{x_1}^{\in \mathbb{R}^4}) \cdots \hat{\phi}(x_n)\big) | 0 \rangle}$$

T → time ordered

$$\boxed{[\hat{\phi}(t,\vec{x}), \hat{\phi}(t,\vec{y})] = 0 \qquad [\hat{q}_i, \hat{q}_j] = 0}$$

$$[\hat{\phi}(t,\vec{x}), \phi(t',\vec{x}')] = 0$$ if $(t,\vec{x})$, $(t',\vec{x}')$ are space like seperated

interactions

$$S[\phi] = \underbrace{\int_{\mathbb{R}^4} d^4x \, \mathcal{L}_{free}(\phi, \partial\phi)}_{S_0[\phi]} + \underbrace{g \int_{\mathbb{R}^4} d^4x \, \mathcal{L}_{int}(\phi, \partial\phi)}_{g \, S_I[\phi]}$$

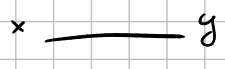
trick $$Z[\phi] = \int D\phi \, e^{iS_0[\phi] + igS_I[\phi]}$$

$$\int d^3x \left(\alpha_3 \phi^3(x) + \alpha_3 \phi^4(x) + \dots\right)$$

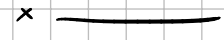
$$= \sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \, e^{iS_0[\phi]} (S_I(\phi))^k$$

$$\langle \phi(x) \phi(y) \rangle = \frac{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \, e^{iS_0[\phi]} (S_I(\phi))^k \, \phi(x) \phi(y)}{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \, e^{iS_0[\phi]} (S_I(\phi))^k}$$

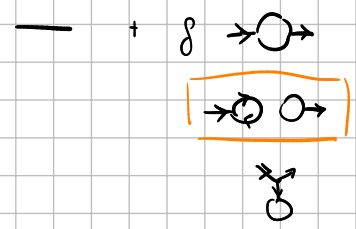
$O(1)$: $\langle \phi(x)\phi(y)\rangle_0$ ← fermion



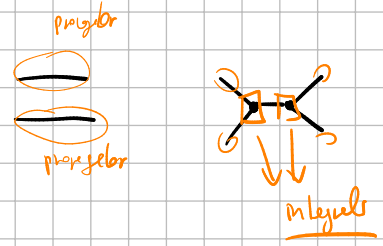
$O(g)$: $\langle \phi(x)\phi(y)\rangle_0$



$O(g^2)$:



<u>3-point</u>



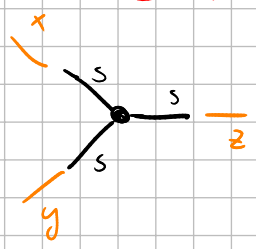
$$\langle \phi(x)\phi(y)\phi(z)\rangle = \frac{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \; e^{iS_0[\phi]} \left(S_I(\phi)\right)^k \phi(x)\phi(y)}{\sum_{k=0}^{\infty} \frac{(ig)^k}{k!} \int D\phi \; e^{iS_0[\phi]} \left(S_I(\phi)\right)^k}$$

expand to $O(g)$

$$= \underbrace{\langle \phi(x)\phi(y)\phi(z)\rangle_0}_{=0} + \boxed{ig \int d^4(s) \; \langle \phi(x)\phi(y)\phi(z)\,\phi^3(s)\rangle_0}$$

$$- ig \underbrace{\langle \phi(x)\phi(y)\phi(z)\rangle_0}_{=0} \int d^4(s) \; \langle \phi^3(s)\rangle_0$$

Something wrong here



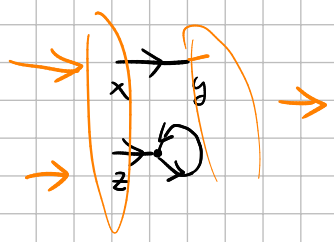
1 2 3 4 5 6 = (12)(34)(56)

(12)(35)(46)

(13)(24)(56)

(13)(25)(46)

(14)(23)



$\langle \phi(x)\phi(y)\rangle \langle \phi(z)\phi(s)\rangle \langle \phi(s)^2\rangle$

$\langle \phi(x)\phi(z)\rangle \langle \phi(y)\phi(s)\rangle \langle \phi(s)^2\rangle$

$\langle \phi(y)\phi(z)\rangle \langle \phi(x)\phi(s)\rangle \langle \phi(s)^2\rangle$

$\langle \phi(x)\phi(s)\rangle \langle \phi(y)\phi(s)\rangle \langle \phi(z)\phi(s)\rangle$

$\langle \phi(x) \rangle =$

<u>momentum space</u>

$$\varphi(p) = \int \frac{d^4x}{(2\pi)^2} \, \phi(x) \, e^{-ip\cdot x}$$

$$\phi(x) = \int \frac{d^4p}{(2\pi)^2} \, \varphi(p) \, e^{+ip\cdot x}$$

$$S[\phi] = -\frac{1}{2}\int d^4x \; \phi \nabla^2 \phi \quad + \; g \int d^4x \; \phi^3(x)$$

$$= \frac{1}{2}\int d^4x \int d^4p_1 \int d^4p_2 \; \frac{1}{(2\pi)^4} \varphi(p_1)\varphi(p_2) \; p_2^2 \; e^{i(p_1+p_2)\cdot x} \quad = (2\pi)^4 \delta(p_1+p_2)$$

$$+ \; g \int d^4(x) \int d^4p_1 \, d^4p_2 \int d^4p_3 \; \frac{1}{(2\pi)^6} \varphi(p_1)\varphi(p_2)\varphi(p_3) \; e^{i(p_1+p_2+p_3)\cdot x} \quad = (2\pi)^4 \delta(p_1+p_2)$$

$$= \frac{1}{2}\int d^4p \left(-p^2 \varphi(p)\varphi(-p)\right) + \frac{g}{(2\pi)^2}\int d^4p_1 \, d^4p_2 \, d^4p_3 \; \varphi(p_1)\varphi(p_2)\varphi(p_3) \; \delta(p_1+p_2+p_3)$$

$$\langle \varphi(p_1) \cdots \varphi(p_k) \rangle = \sum_{l=0}^{\infty} g^k \int D\varphi \; e^{-S_0[\varphi]} \; S_I([\varphi]) \; \varphi(p_1) \cdots \varphi(p_k)$$

$$\langle \varphi(p_1)\, \varphi(p_2) \rangle_0 = \frac{1}{(p_1+p_2)^2}$$

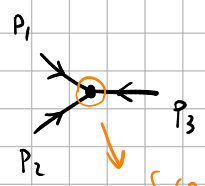
s_{12}

$$\frac{1}{2}\int d^4p \left(-p^2 \varphi(p)\varphi(-p)\right)$$

$$\downarrow$$

$$\frac{1}{2}\int d^4p_1 \, d^4p_2 \left(-p_1\cdot p_2 \; \varphi(p_1)\varphi(p_2)\right) \delta(p_1+p_2)$$

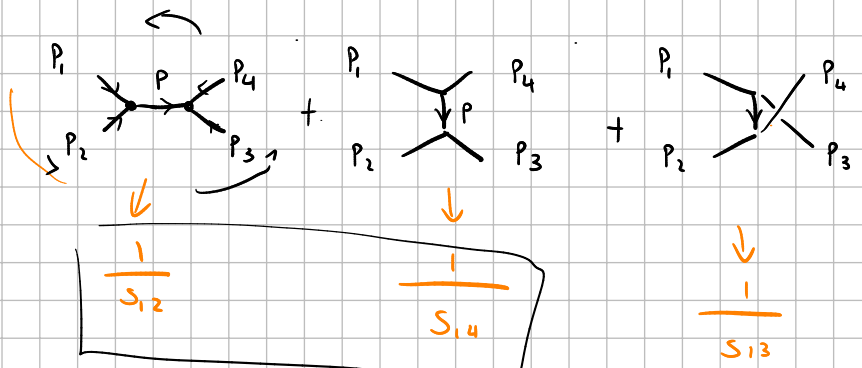
$s_{ij} = (p_i + p_j)^2$

$\langle \varphi(p_1)\, \varphi(p_2)\, \varphi(p_3) \rangle =$ 

$+ \; O(g^2)$

$\delta(p_1+p_2+p_3)\, g$

$\langle \varphi(p_1)\,\varphi(p_2)\,\varphi(p_3)\,\varphi(p_4)\rangle =$



sum over permutations of only non-cherry ones

$$2\left(\frac{1}{s_{23}}+\frac{1}{s_{34}}\right)+\left(\frac{1}{s_{12}}+\frac{1}{s_{14}}\right)+\left(\frac{1}{s_{12}}+\frac{1}{s_{13}}\right)+\left(\frac{1}{s_{24}}+\frac{1}{s_{14}}\right)+\left(\frac{1}{s_{34}}+\frac{1}{s_{14}}\right)$$

$$2\left(\frac{1}{12}\,\frac{1}{14}\right)+\left(\frac{1}{12}\,\frac{1}{13}\right)+\left(\frac{1}{14}\,\frac{1}{13}\right)$$

$$8\,\frac{1}{s_{12}}+8\,\frac{1}{s_{14}}+4\,\frac{1}{s_{13}}$$

---

eq 2

$$\mathcal{A}_n = \sum_{\sigma\in S_n/Z_n} A_n(\sigma(1),\dots,\sigma(n))$$

ordered

1 2 3 4 ~ 2 3 4 1

$$A_4(1,2,3,4) = \frac{1}{s_{12}}+\frac{1}{s_{14}}$$

12	14
12	13
13	12
13	14
14	13
14	12

$$4\left(\frac{1}{s_{12}}+\frac{1}{s_{13}}+\frac{1}{s_{14}}\right)$$

✓ 1 2 3 4
1 2 4 3
1 3 4 2
1 3 2 4
1 4 2 3
1 4 3 2

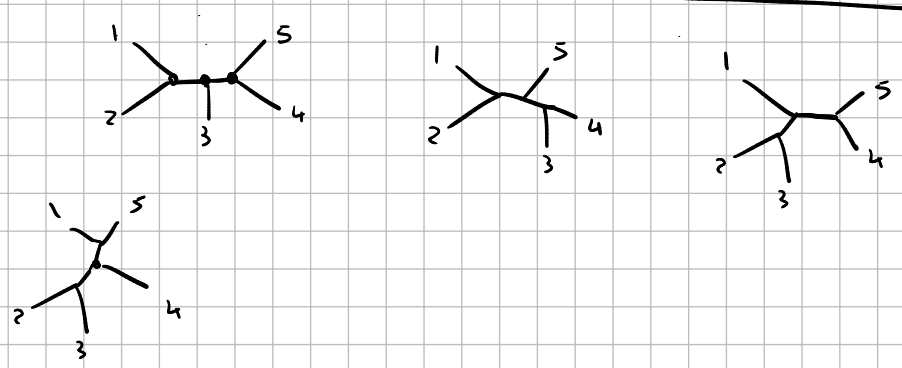
~~2 1 3 4~~
~~2 1 4 3~~
~~2 3 1 4~~
~~2 3 4 1~~
~~2 4 1 3~~
~~2 4 3 1~~

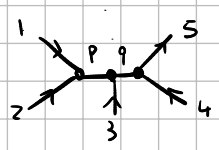
~~3 1 2 4~~
~~3 1 4 2~~
~~3 2 1 4~~
~~3 2 4 1~~
~~3 4 1 2~~
~~3 4 2 1~~

~~4 1 2 3~~
~~4 1 3 2~~
~~4 2 1 3~~
~~4 2 3 1~~
~~4 3 1 2~~
~~4 3 2 1~~

---

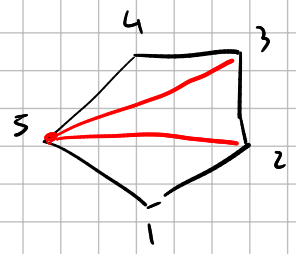
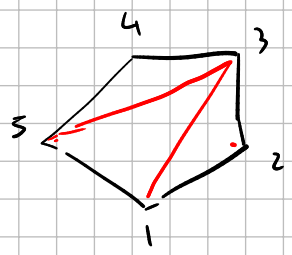
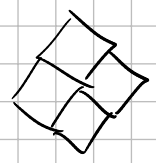
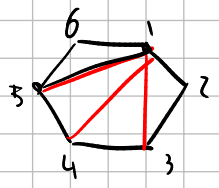
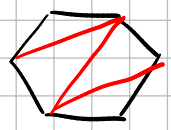
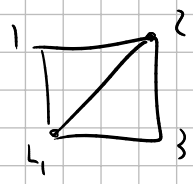
5 point





$$\int d^3q \int d^3p \, \frac{1}{p^2} \, \delta(p_1+p_2-p) \, \delta(p-q+p_3) \, \frac{1}{q^2} \, \delta(p+p_3-q) \, \delta(q+p_5+p_4)$$

$$\frac{1}{(p_1+p_2)^2} \, \frac{1}{(p_4+p_5)^2} = \frac{\delta(p_1+p_2+p_3+p_4)}{s_{12}\, s_{45}}$$

$$\frac{1}{s_{12}\, s_{45}} + \frac{1}{s_{12}\, s_{34}} + \frac{1}{s_{23}\, s_{45}} + \frac{1}{s_{23}\, s_{15}} + \frac{1}{\boxed{s_{34}}\, s_{15}}$$

$$s_{ij} = (p_i + p_j)^2 = 2 p_i \cdot p_j$$